

DATA 260 — Homework 1

Haozhe Wang

SID4 = 8561

PORT_BASE = 8461

PREFIX = s8561

SEED = 8561

VERIFY_SEED = 268561

DOMAIN_ID = 1 (Clinical Trial listings)

Hardware: MacBook Pro 14", Nov 2024(Apple M4, 16 GB unified memory, macOS Tahoe 26.3)

Local model: qwen3:8b via Ollama

Python: 3.12.14

Repository: <https://github.com/HauZer-HaoZheWang/data260-8561>

Tagged commit (hw1): 7060a5422f2c84e68c3764b788f92ae18898f685

Questions: (Part 1)

===== index.html =====

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <title>HW1-Haozhe Wang</title>
```

```
</head>
```

```
<body>
```

```
  <h1>Clinical Trial Registration</h1>
```

```
  <form id="trialForm">
```

```
    <label for="briefTitle">Brief Title</label>
```

```
    <input type="text" id="briefTitle" name="briefTitle"
```

```
      placeholder="Please type in the brief title of trial description which can be easily read by people."
      required autofocus>
```

```

<label for="sponsor">Sponsor</label>
<input type="text" id="sponsor" name="sponsor"
  placeholder="Please type in the sponsor of trial, e.g. University of Colorado, Denver."
  required>

<label for="submitterEmail">Submitter Email</label>
<input type="email" id="submitterEmail" name="submitterEmail"
  placeholder="email_address_of_submitter@trial.com" required>

<label for="detailedDescription">Detailed Description</label>
<textarea id="detailedDescription" name="detailedDescription" rows="6"
  placeholder="Write at least 25 characters to describe the details of trial."
  required></textarea>

<label for="funderType">Funder Type</label>
<select id="funderType" name="funderType" required>
  <option value="">-- Select a funder type --</option>
  <option value="NIH">NIH</option>
  <option value="Other U.S. federal agency">Other U.S. federal agency</option>
  <option value="Industry">Industry</option>
  <option value="All others (individuals, universities, organizations)">All others (individuals, universities,
organizations)</option>
</select>

<input type="checkbox" id="agreeTerms" name="agreeTerms" required>
<label for="agreeTerms">I agree to the terms and conditions.</label>

<button type="submit">Submit Trial!</button>
</form>

<script src="feedback.js"></script>
</body>
</html>

```

===== feedback.js =====

```

function makeSubmission() {
  let count = 0;
  return function () {
    count = count + 1;
    return count;
  };
}

```

```
const countSubmission = makeSubmission();
```

```
const trialForm = document.getElementById("trialForm");
```

```
trialForm.addEventListener("submit", (event) => {
```

```
  event.preventDefault();
```

```

  const detailedDescription = document.getElementById("detailedDescription").value;
  if (detailedDescription.length <= 25) {
    alert("Your detailed description must more than 25 characters.");
    return;
  }
}

```

```
const agreeTerms = document.getElementById("agreeTerms").checked;
if (!agreeTerms) {
  alert("Please check 'I agree to the terms and conditions.'");
  return;
}

const trialData = Object.fromEntries(new FormData(trialForm));
const jsonString = JSON.stringify(trialData);
console.log("Submitted JSON:", jsonString);

const parsed = JSON.parse(jsonString);
const { briefTitle, submitterEmail } = parsed;
console.log("Brief Title:", briefTitle);
console.log("Submitter Email:", submitterEmail);

const withDate = { ...parsed, submissionDate: new Date().toISOString() };
console.log("With date:", withDate);

const total = countSubmission();
console.log("Submission count:", total);
});
```

```
===== Dockerfile =====
FROM nginx:alpine
```

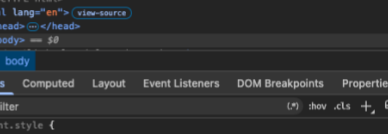
```
RUN rm -rf /usr/share/nginx/html/*
```

```
COPY web_application/ /usr/share/nginx/html/
```

```
EXPOSE 80
```

```
CMD ["nginx", "-g", "daemon off;"]
```

Clinical Trial Registration



```

[?] Console AI assistance
ID  top  [refresh]  Filter  Default levels  No issues  [refresh]

With date: feedback:js:41
> {urlTitle: '12345', sponsor: 'aa35123451234512345123456',
  $loot.com, detailedDescription: 'hazoe.wang@jssu.edu',
  type: 'NIH', -}

Submission count: 3 feedback:js:44
Submitted 350M: {"urlTitle": "A brief title of a
  clinical trial.", "sponsor": "Big
  Boss", "submitterEmail": "hazoe.wang@jssu.edu", "detailedDescription": "This
  is a detailed description of a clinical
  trial.", "funderType": "NIH", "agreements": "on"}
  feedback:js:33
Brief Title: A brief title of a clinical trial. feedback:js:37
Submitter Email: hazoe.wang@jssu.edu feedback:js:38
With date: feedback:js:41
> {urlTitle: 'A brief title of a clinical trial.', sponsor: 'Big Bos
  s', submitterEmail: 'hazoe.wang@jssu.edu', detailedDescription: 'This
  is a detailed description of a clinical trial.', funderType: 'NIH', -}
Submission count: 4 feedback:js:44
Submitted 350M: {"urlTitle": "A brief title of a
  clinical trial.", "sponsor": "Big
  Boss", "submitterEmail": "hazoe.wang@jssu.edu", "detailedDescription": "This
  is a detailed description of a clinical
  trial.", "funderType": "NIH", "agreements": "on"}
  feedback:js:33
Brief Title: A brief title of a clinical trial. feedback:js:37
Submitter Email: hazoe.wang@jssu.edu feedback:js:38
With date: feedback:js:41
> {urlTitle: 'A brief title of a clinical trial.', sponsor: 'Big Bos
  s', submitterEmail: 'hazoe.wang@jssu.edu', detailedDescription: 'This
  is a detailed description of a clinical trial.', funderType: 'NIH', -}
Submission count: 5 feedback:js:44
Submitted 350M: {"urlTitle": "A brief title of a
  clinical trial.", "sponsor": "Big
  Boss", "submitterEmail": "hazoe.wang@jssu.edu", "detailedDescription": "This
  is a detailed description of a clinical
  trial.", "funderType": "NIH", "agreements": "on"}
  feedback:js:33
Brief Title: A brief title of a clinical trial. feedback:js:37
Submitter Email: hazoe.wang@jssu.edu feedback:js:38
With date: feedback:js:41
> {urlTitle: 'A brief title of a clinical trial.', sponsor: 'Big Bos
  s', submitterEmail: 'hazoe.wang@jssu.edu', detailedDescription: 'This
  is a detailed description of a clinical trial.', funderType: 'NIH', -}
Submission count: 6 feedback:js:44
> [cmd] to turn on code suggestions. Don't show again [refresh]
```

us-west-1.console.aws.amazon.com

Amazon Web Services Sign-in

Task definition containers | Elastic Container Service | us-west-1

Search

[Option+S]

United States (N. California)

Haozhe (877710816843)

Haozhe

Amazon Elastic Container Service

Task definitions

s8561-task

Revision 1

Containers

Amazon Elastic Container Service

Favorites and recents

Express Mode

Clusters

Namespaces

Task definitions

Daemon task definitions

Account settings

AWS Resource Explorer

AWS Batch

Amazon ECR

Repositories

Online learning workshop

Documentation

Discover products

Subscriptions

Tell us what you think

s8561-task:1

Last updated August 31, 2026, 04:06 (UTC-7:00)

Deploy

Actions

Create new revision

Overview

Info

ARN

am:aws:ecs:us-west-1:877710816843:task-definition/s8561-task:1

Status

ACTIVE

Time created

August 31, 2026, 04:06 (UTC-7:00)

App environment

Fargate

Task role

-

Task execution role

ecsTaskExecutionRole

Operating system/Architecture

Linux/ARM64

Network mode

awsvpc

Fault injection

Turned off

Containers

JSON

Task placement

Volumes (0)

Requires attributes

Tags

Task size

Task CPU

256 units (0.25 vCPU)

Task CPU maximum allocation for containers

CPU (unit)

0 20 40 60 80 100 120 140 160 180 200 220 240

s8561-web Shared task CPU

Task memory

512 MiB (0.5 GiB)

Task memory maximum allocation for container memory reservation

Memory (MiB)

0 50 100 150 200 250 300 350 400 450 500

s8561-web Shared task memory

Container: s8561-web

Info

Essential container

Details

JSON

Image

877710816843.dkr.ecr.us-west-1.amazonaws.com/s8561-hw1:latest

Private registry

Turned off

Secrets Manager ARN or name

-

CPU

0

Memory hard/soft limit

-/-

GPU

-

Neuron device

-

Environment and secrets

Network settings

Security and permissions

Lifecycle and dependencies

Monitor

Environment variables (0)

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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Privacy

Terms

Cookie preferences

Clinical Trial Registration

Brief Title

Please type in the brief title

Sponsor

Please type in the sponsor

Submitter Email

email_address_of_submitt

Detailed Description

Write at least 25 characters to describe the details of trial.

Funder Type

-- Select a funder type --

☐ I agree to the terms and conditions.

Submit Trial!

Agentic AI (Part 2)

===== PART 2 OUTPUT =====

--- Planner (21958 ms) ---

```
{  
  "thought": "Identifying key themes in the study's focus on DBS, local field potentials, and motor symptom evaluation.",  
  "message": "Tags and summary created based on the study's focus on DBS, local field potentials, and motor symptom evaluation in neurological disorders.",  
  "data": {  
    "tags": [  
      "Deep Brain Stimulation Research",  
      "Local Field Potentials Analysis",  
      "Motor Symptom Evaluation in Neurological Disorders"  
    ],  
    "summary": "This study examines motor symptom responses in Parkinson's and related disorders using local field potentials from deep brain stimulation electrodes.",  
    "issues": []  
  }  
}
```

--- Reviewer (42261 ms) ---

```
{  
  "thought": "Tags focus on study methods and conditions; summary captures key objective.",  
  "message": "Tags and summary created based on study's focus on DBS, local field potentials, and motor symptom evaluation in neurological disorders.",
```

```
"data": {
  "tags": [
    "Deep Brain Stimulation",
    "Local Field Potentials",
    "Motor Symptom Evaluation"
  ],
  "summary": "Study examines DBS's impact on motor symptoms via local field potentials in Parkinson's and movement disorders.",
  "issues": []
}
```

Finalized Output

```
{
  "thought": "Finalizing tags and summary based on study's focus on DBS, local field potentials, and motor symptom evaluation in neurological disorders.",
  "message": "Tags and summary refined to reflect study's core focus on DBS, neural recordings, and motor symptom assessment in movement disorders.",
  "data": {
    "tags": [
      "Deep Brain Stimulation",
      "Local Field Potentials",
      "Neurological Disorders"
    ],
    "summary": "This study examines motor symptom responses in Parkinson's and related disorders using DBS electrode recordings of brain activity.",
    "issues": []
  }
}
```

```
}  
}
```

Publish Package

```
{  
  "title": "Deep Brain Stimulation for Parkinson's Disease",  
  "email": "student@example.com",  
  "content": "This observational study records local field potentials from deep brain  
stimulating electrodes implanted in patients with Parkinson disease, essential tremor, or  
dystonia, to evaluate motor symptom response.",  
  "agents": {  
    "transcript": [  
      {  
        "role": "Planner",  
        "content": "Tags and summary created based on the study's focus on DBS, local field  
potentials, and motor symptom evaluation in neurological disorders."  
      },  
      {  
        "role": "Reviewer",  
        "content": "Tags and summary created based on study's focus on DBS, local field  
potentials, and motor symptom evaluation in neurological disorders."  
      }  
    ],  
    "final": {  
      "tags": [  
        "Deep Brain Stimulation",  
        "Local Field Potentials",
```

```
"Neurological Disorders"

],

"summary": "This study examines motor symptom responses in Parkinson's and related
disorders using DBS electrode recordings of brain activity.",

"issues": []

}

},

"submissionDate": "2026-08-31T06:51:38Z"

}
```

Command used:

```
python code/agents_demo.py --title "Deep Brain Stimulation for Parkinson's Disease" --
content "This observational study records local field potentials from deep brain stimulating
electrodes implanted in patients with Parkinson disease, essential tremor, or dystonia, to
evaluate motor symptom response."
```

Q1. Final tags:

Deep Brain Stimulation, Local Field Potentials, Neurological Disorders

Q2. Final summary;

"This study examines motor symptom responses in Parkinson's and related disorders using DBS electrode recordings of brain activity."

Q3. Did the Reviewer change anything? Yes.

The Planner proposed tags with redundant suffixes: "Deep Brain Stimulation Research", "Local Field Potentials Analysis", "Motor Symptom Evaluation in Neurological Disorders". The Reviewer stripped the Research/Analysis suffixes and shortened the third tag. The

Finalizer then replaced "Motor Symptom Evaluation" with the broader "Neurological Disorders" and rewrote the summary from 21 to 20 words.

Explanation of each step:

The Planner reads the title and content and proposes three tags plus a one-line summary. The Reviewer checks whether the tags are topical rather than generic and whether the summary stays within 25 words, recording any problems in data.issues. The Finalizer applies the reviewer's feedback and emits the final strict-JSON output. All three agents call the same qwen3:8b model and differ only in their system prompt. The domain is never hardcoded: tags are derived from the input title and content, so the same pipeline works on any domain.

Measuring Non-Determinism (Part 3)

Fixed input (reports/hw01/cases/nondeterminism_input.json):

```
{  
  "title": "Deep Brain Stimulation for Parkinson's Disease",  
  "content": "This observational study records local field potentials from deep brain stimulating electrodes implanted in patients with Parkinson disease, essential tremor, or dystonia, to evaluate motor symptom response."  
}
```

Part 3 Results

Temp 0.7:

Distinct tag sets: 13

Tags in all 20 runs: Deep Brain Stimulation

Tags in exactly 1 run: 9

Latency p50/p95/p99: 92533 / 153022 / 180020 ms

Temp 0.0:

Distinct tag sets: 1

Tags in all 20 runs: Deep Brain Stimulation, Local Field Potentials, Neurological Disorders

Tags in exactly 1 run: 0

Latency p50/p95/p99: 78362 / 81361 / 84505 ms

At temperature 0.7 the pipeline produced 13 different tag sets across 20 runs. Only "Deep Brain Stimulation" appeared in every run, and 9 tags appeared exactly once. The variation was almost entirely in near-synonyms: Movement Disorders / Neurological Disorders / Neurological Conditions, and Local Field Potentials / Electrophysiological Recordings / Neurophysiological Data. The model was stable about what the text is about, but unstable about which words to use.

At temperature 0.0 all 20 runs produced an identical tag set and latency varied by under 7 percent. In this configuration greedy decoding was fully reproducible. This is an empirical result, not a guarantee: reproducibility at temperature 0 depends on hardware, batching and floating-point reduction order, so a different machine or backend could still show variation.

Two users sending identical input at temperature 0.7 would most likely see different tag sets: only 5 of 20 runs produced the most common set. At temperature 0.0 they would see identical output.

Run-to-run variation is acceptable when the output is a suggestion a human review before use, such as suggested tags on a draft post, where near-synonyms carry the same meaning.

Run-to-run variation is not acceptable when the output feeds an automated decision or must be auditable, such as a drug dosage calculation or a regulatory filing, where two

identical submissions must produce identical records.

```
def pct(sorted_vals, p):
    if not sorted_vals:
        return 0
    k = (len(sorted_vals) - 1) * p / 100
    lo, hi = int(k), min(int(k) + 1, len(sorted_vals) - 1)
    return round(sorted_vals[lo] + (sorted_vals[hi] - sorted_vals[lo]) * (k - lo))

for temp in [0.7, 0.0]:
    rows = [r for r in data if r["temperature"] == temp and r["tags"]]
    sets = [tuple(sorted(r["tags"])) for r in rows]
    tag_counts = Counter(t for s in sets for t in set(s))
    lat = sorted(r["latency_ms"] for r in rows)

    print(f"\n--- temperature = {temp} (n={len(rows)}) ---")
    print("Distinct tag sets      :", len(set(sets)))
    print("Tags in ALL runs          :", [t for t, c in tag_counts.items() if c == len(rows)])
    print("Tags in EXACTLY 1 run      :", [t for t, c in tag_counts.items() if c == 1])
    print(f"Latency p50/p95/p99 ms : {pct(lat, 50)} / {pct(lat, 95)} / {pct(lat, 99)}")
    print("Unique sets detail:")
    for s, c in Counter(sets).most_common():
        print(f"    {c}x {list(s)}")

PYEOF
python3 code/analyze_nondeterminism.py | tee reports/hw01/part3_stats.txt

--- temperature = 0.7 (n=20) ---
Distinct tag sets      : 13
Tags in ALL runs       : ['Deep Brain Stimulation']
Tags in EXACTLY 1 run  : ['Motor Response Analysis', 'Neurological Conditions', 'Neurophysiological Monitoring', 'Motor Symptoms', 'Movement Disorder Assessment', 'Neurophysiological Data', 'Electrophysiological Recordings', 'Motor Symptom Evaluation', 'Neurological Movement Disorders']
Latency p50/p95/p99 ms : 92533 / 153022 / 180020
Unique sets detail:
5x ['Deep Brain Stimulation', 'Local Field Potentials', 'Movement Disorders']
2x ['Deep Brain Stimulation', 'Local Field Potentials', 'Neurological Disorders']
2x ['Deep Brain Stimulation', 'Movement Disorders', 'Neural Signal Analysis']
2x ['Deep Brain Stimulation', 'Local Field Potentials', 'Motor Symptom Assessment']
1x ['Deep Brain Stimulation', 'Local Field Potentials', 'Motor Response Analysis']
1x ['Deep Brain Stimulation', 'Electrophysiological Monitoring', 'Movement Disorders']
1x ['Deep Brain Stimulation', 'Local Field Potentials', 'Neurological Conditions']
1x ['Deep Brain Stimulation', 'Motor Symptoms', 'Neurophysiological Monitoring']
1x ['Deep Brain Stimulation', 'Movement Disorder Assessment', 'Neurophysiological Data']
1x ['Deep Brain Stimulation', 'Electrophysiological Recordings', 'Movement Disorders']
1x ['Deep Brain Stimulation', 'Local Field Potentials', 'Motor Symptom Evaluation']
1x ['Deep Brain Stimulation', 'Local Field Potentials', 'Neurological Movement Disorders']
1x ['Deep Brain Stimulation', 'Electrophysiological Monitoring', 'Neurological Disorders']

--- temperature = 0.0 (n=20) ---
Distinct tag sets      : 1
Tags in ALL runs       : ['Local Field Potentials', 'Deep Brain Stimulation', 'Neurological Disorders']
Tags in EXACTLY 1 run  : []
Latency p50/p95/p99 ms : 78362 / 81361 / 84505
Unique sets detail:
20x ['Deep Brain Stimulation', 'Local Field Potentials', 'Neurological Disorders']
(data260) haozhe@haozheMacBook-Pro data260-8561 %
```

Model Client and Token Accounting (Part 4)

```
> Review this: def add(a, b): return a - b
Review this: def div(a, b): return a / b
```

- Function name 'add' does not match the subtraction operation.
- The operation should be addition (a + b) instead of subtraction.
- No error handling for invalid input types.
- The function lacks docstring for clarity.
- The code is overly simplified and does not follow standard practices.
- The function could be renamed to 'subtract' for consistency.

```
[turn 1] input=112 output=493 total=605
```

- ```
>
```
- Function lacks error handling for division by zero.
  - No docstring explaining the purpose and parameters.
  - Division operator '/' may perform integer division in Python 2.
  - The function could benefit from type hints for clarity.
  - No input validation for non-numeric arguments.
  - The function name 'div' is correct but could be more descriptive.

```
[turn 2] input=212 output=522 total=734
```

```
> Review this: x = open("data.txt").read()
```

- The code is invalid due to the "/think" suffix.
- No error handling for file not found or read errors.
- Using open without a context manager (with statement) leads to resource leaks.
- No encoding specified for the file, which may cause issues with non-UTF-32 files.
- The variable x is assigned the entire file content as a string, which may not be necessary.
- The code lacks comments or docstrings explaining the purpose.

```
[turn 3] input=308 output=900 total=1208
```

```
> /stats
--- STATS ---
turns : 3
cumulative input tokens : 632
cumulative output tokens : 1915
serialized history length : 411 (estimated)

```

```
> Review this: for i in range(len(items)): print(items[i])
```

- The code uses index-based iteration instead of directly iterating over the list for simplicity.
- No error handling for empty 'items' or non-iterable types.
- The variable 'i' is less descriptive than using 'index' for clarity.
- The code could be simplified by directly iterating over 'items' instead of using range(len()).
- No input validation to prevent potential NoneType errors.
- The code lacks comments or docstrings explaining the purpose.

```
[turn 4] input=431 output=467 total=898
```

```
> Review this: password = "admin123"
```

- Hardcoded password in plain text violates security best practices.
- Password 'admin123' is weak and easily guessable.
- No encryption or hashing applied to the password.
- Variable name 'password' is acceptable but could be more descriptive.
- The code lacks input validation or security checks.
- Suggest using environment variables or a secrets manager for sensitive credentials.

```

[turn 3] input=308 output=900 total=1208

> /stats
--- STATS ---
turns : 3
cumulative input tokens : 632
cumulative output tokens : 1915
serialized history length : 411 (estimated)

> Review this: for i in range(len(items)): print(items[i])

- The code uses index-based iteration instead of directly iterating over the list for simplicity.
- No error handling for empty 'items' or non-iterable types.
- The variable 'i' is less descriptive than using 'index' for clarity.
- The code could be simplified by directly iterating over 'items' instead of using range(len()).
- No input validation to prevent potential NoneType errors.
- The code lacks comments or docstrings explaining the purpose.

[turn 4] input=431 output=467 total=898

> Review this: password = "admin123"

- Hardcoded password in plain text violates security best practices.
- Password 'admin123' is weak and easily guessable.
- No encryption or hashing applied to the password.
- Variable name 'password' is acceptable but could be more descriptive.
- The code lacks input validation or security checks.
- Suggest using environment variables or a secrets manager for sensitive credentials.

[turn 5] input=549 output=561 total=1110

> /stats
--- STATS ---
turns : 5
cumulative input tokens : 1612
cumulative output tokens : 2943
serialized history length : 647 (estimated)

> █

```

Q1. Why is prior conversation context resent with every turn?

The model is stateless. Each call sees only the messages passed in that call and retains nothing between calls. The apparent memory is maintained entirely by the client, which appends each user and assistant message to a list and resends the whole list every turn.

Q2. How is a system prompt different from a user message?

The system prompt defines role and persistent constraints, sits first in the message list, and is resent every turn, so it applies to the whole session. AGENT.md is the example here: it constrains how the model answers. A user message is a single-turn request specifying what to answer.

Q3. Why do input tokens grow over a conversation?

Because the full history is resent each turn and the history only grows. Measured here: 112, 212, 308, 431, 549 input tokens across five turns. The increment itself grows, so cumulative input cost is quadratic while the history length is linear. At turn 5 cumulative input was 1612 against a history length of 647.

Q4. What eventually limits that growth?

The context window, the maximum number of tokens the model can process in one call. Beyond it the request fails or the oldest messages are silently dropped. Mitigations include sliding-window truncation, summarizing older turns, and retrieving from an external memory store on demand.

Full code, logs, raw data, METRICS.md, AI\_USE.md, DOMAIN\_SCHEMA.md and verification.json:

see tagged commit hw1 at <https://github.com/HauZer-HaoZheWang/data260-8561>